

Comprehensive Air Quality Assessment Report

Hay Hassani Monitoring Station, Casablanca - 2014

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Executive Summary

This report presents a comprehensive analysis of air quality data from the Hay Hassani monitoring station in Casablanca for the calendar year 2014. The assessment reveals significant insights into pollution patterns, meteorological influences, and regulatory compliance status according to Moroccan environmental standards (Decree 2-09-286 of December 8, 2009). Primary pollutants such as NO₂, SO₂, and CO comply with standards, while PM₁₀ and O₃ exhibit persistent exceedances, requiring targeted corrective measures.

1 Introduction

1.1 Study Framework

The Hay Hassani monitoring station represents an urban background site in Casablanca, providing critical data on atmospheric pollution levels. This analysis covers the full 2014 dataset, comprising 8,760 hourly measurements of multiple pollutants (NO₂, O₃, PM₁₀, SO₂, CO) and meteorological parameters (wind direction and speed, temperature, humidity). The analyzed period spans from January 1, 2014, at 00:00 to January 1, 2015, at 00:00, covering 366 days.

1.2 Data Coverage and Quality

The dataset demonstrates excellent completeness, with coverage exceeding 87% for all measured parameters. Data quality is rated as good to excellent across all pollutants.

Table 1: Data coverage by pollutant

Pollutant	Coverage (%)	Evaluation
NO ₂	99.0	Excellent
O ₃	94.7	Good
PM ₁₀	87.2	Good
SO ₂	96.3	Excellent
CO	93.8	Good

2 Regulatory Compliance

Standards are established by Decree No. 2-09-286 of December 8, 2009, which defines limit values for the protection of human health and vegetation.

2.1 Overall Compliance Status

The station demonstrates mixed compliance with Moroccan air quality standards. Primary gaseous pollutants are compliant, but particulate matter and ozone show non-compliances.

Table 2: Regulatory compliance status

Pollutant	Status	Observations
NO ₂	COMPLIANT	Meets all limits
SO ₂	COMPLIANT	Levels well below limits
PM ₁₀	NON-COMPLIANT	Exceeds 90.4th percentile daily
O ₃	NON-COMPLIANT	Exceeds 8-hour averages and consecutive days
CO	COMPLIANT	Levels below limits

2.2 Detailed Compliance Results

Table 3: Detailed compliance results

Pollutant	Parameter	Value	Limit	Status
NO ₂	Annual mean	18.9 $\mu\text{g m}^{-3}$	50 $\mu\text{g m}^{-3}$	COMPLIANT
	98th percentile hourly	71.2 $\mu\text{g m}^{-3}$	200 $\mu\text{g m}^{-3}$	COMPLIANT
SO ₂	Annual mean	4.0 $\mu\text{g m}^{-3}$	20 $\mu\text{g m}^{-3}$	COMPLIANT
	99.2th percentile daily	17.5 $\mu\text{g m}^{-3}$	125 $\mu\text{g m}^{-3}$	COMPLIANT
PM ₁₀	90.4th percentile daily	62.7 $\mu\text{g m}^{-3}$	50 $\mu\text{g m}^{-3}$	NON-COMPLIANT
CO	Maximum daily 8h	2.68 mg m^{-3}	10 mg m^{-3}	COMPLIANT
O ₃	Maximum 8h	110.2 $\mu\text{g m}^{-3}$	110 $\mu\text{g m}^{-3}$	NON-COMPLIANT
	Consecutive days >65	12 days	3 days	NON-COMPLIANT

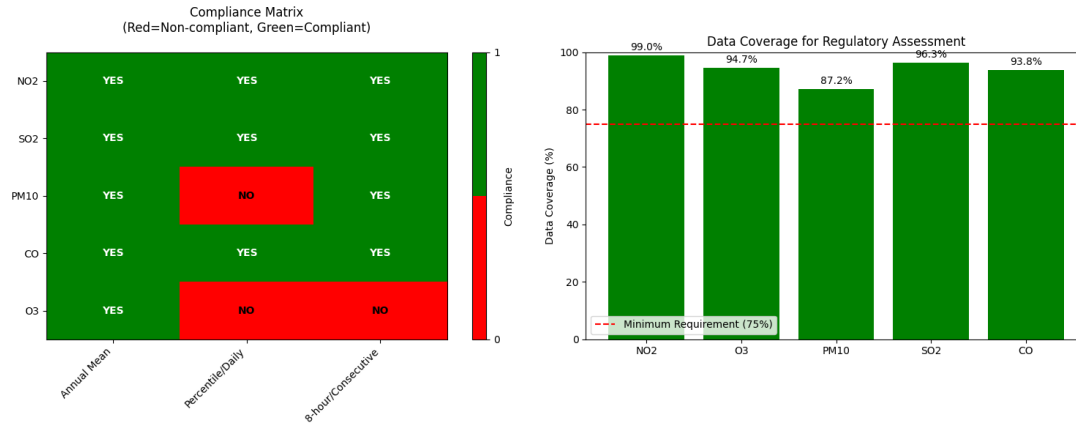


Figure 1: Compliance matrix (Red: Non-compliant, Green: Compliant) and data coverage for regulatory assessment.

Percentile calculations use the linear interpolation method recommended by WHO, with a minimum 75% data completeness requirement, which is met for all pollutants.

3 Pollutant Concentration Analysis

3.1 Descriptive Statistics

Table 4: Annual pollutant statistics

Pollutant	Mean	Maximum	95th Percentile	Exceedances
NO ₂	18.9 $\mu\text{g m}^{-3}$	157.7 $\mu\text{g m}^{-3}$	55.3 $\mu\text{g m}^{-3}$	1,064 hours (>40)
O ₃	50.6 $\mu\text{g m}^{-3}$	149.6 $\mu\text{g m}^{-3}$	93.6 $\mu\text{g m}^{-3}$	140 hours (>100)
PM ₁₀	33.8 $\mu\text{g m}^{-3}$	324.1 $\mu\text{g m}^{-3}$	76.5 $\mu\text{g m}^{-3}$	1,612 hours (>45)
SO ₂	4.0 $\mu\text{g m}^{-3}$	83.1 $\mu\text{g m}^{-3}$	11.2 $\mu\text{g m}^{-3}$	123 hours (>20)
CO	0.787 mg m^{-3}	3.894 mg m^{-3}	1.542 mg m^{-3}	0 hours (>10)

Annual mean concentrations are relatively low for gaseous pollutants, but maximum values indicate episodic high pollution, likely linked to local sources or adverse meteorological conditions.

3.2 Temporal Trends

Monthly trends show clear seasonal variations: NO₂ and PM₁₀ are higher in winter and autumn, while O₃ peaks in spring and summer due to photochemical formation.

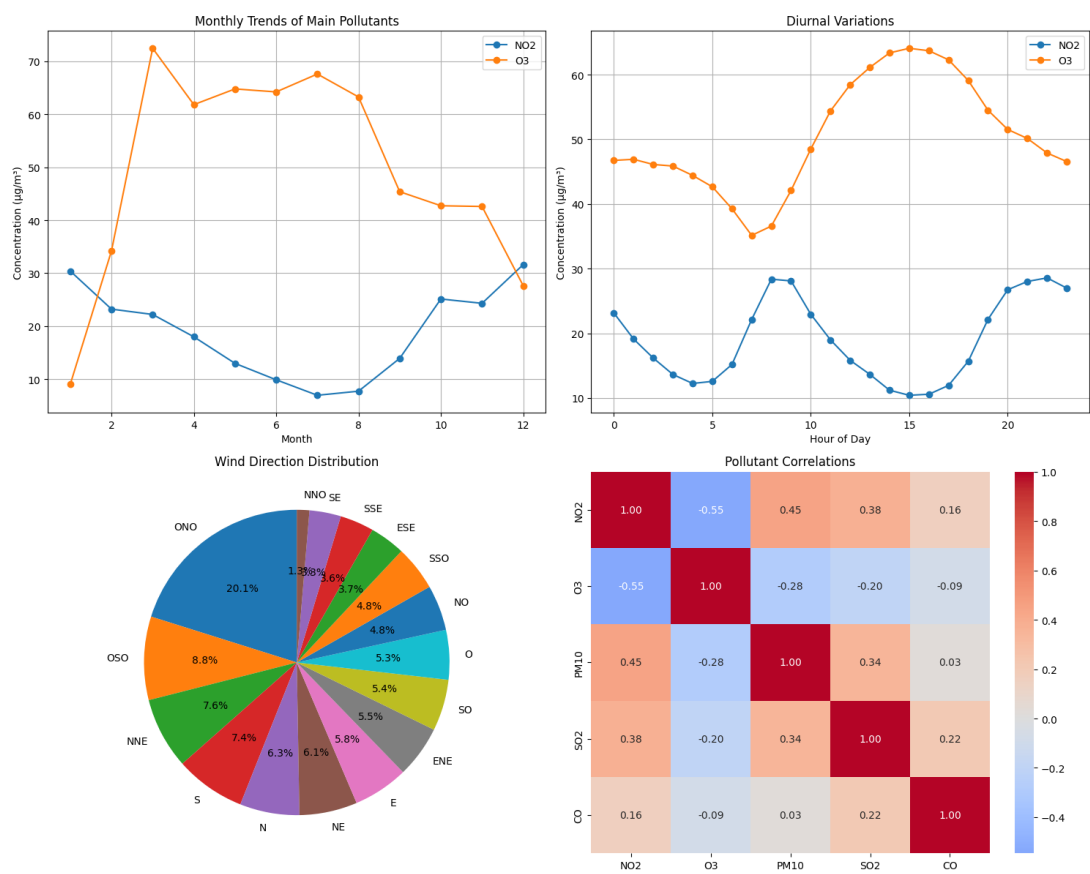


Figure 2: Monthly trends of key pollutants (NO₂, O₃), diurnal variations (NO₂, O₃), wind direction distribution, and pollutant correlations.

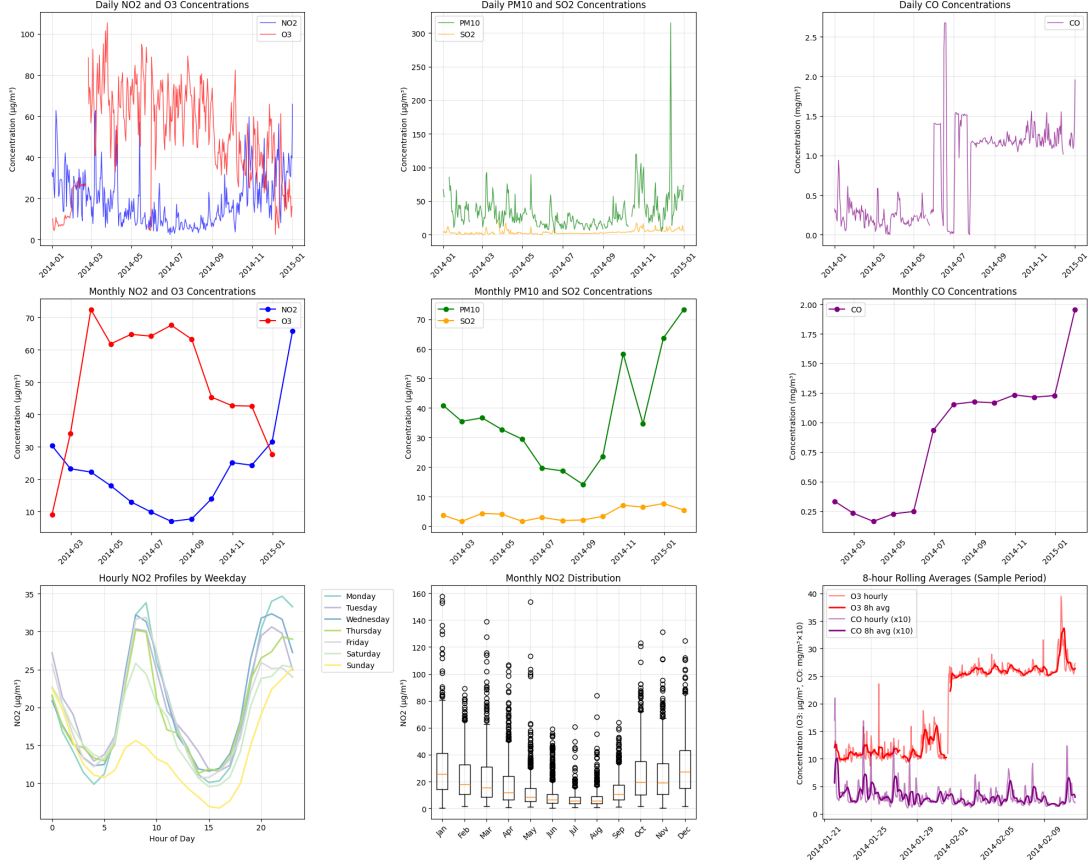


Figure 3: Daily concentrations of NO₂ and O₃, PM₁₀ and SO₂, CO; monthly concentrations of NO₂ and O₃, PM₁₀ and SO₂, CO; hourly NO₂ profiles by weekday; monthly NO₂ distribution; 8-hour rolling averages (sample period).

Diurnal profiles reveal morning and evening peaks for NO₂, corresponding to traffic rush hours, with a maximum weekday-weekend difference of 11.8 g/m³ at 9:00 AM.

3.3 Seasonal and Diurnal Profiles

Table 5: Seasonal concentration variations

Pollutant	Winter	Spring	Summer	Autumn
NO ₂	High (28.1)	Moderate (17.7)	Low (8.9)	Moderate (27.0)
O ₃	Low (23.6)	High (66.4)	High (65.2)	Moderate (43.6)
PM ₁₀	High (46.7)	Moderate (32.9)	Low (17.5)	High (42.2)
SO ₂	Moderate (4.3)	Moderate (4.1)	Low (3.2)	Moderate (4.5)
CO	High (0.92)	Moderate (0.75)	Low (0.65)	Moderate (0.88)

Seasonal variations indicate meteorological influence: winter thermal inversions favor primary pollutant accumulation, while high summer solar radiation promotes ozone formation.

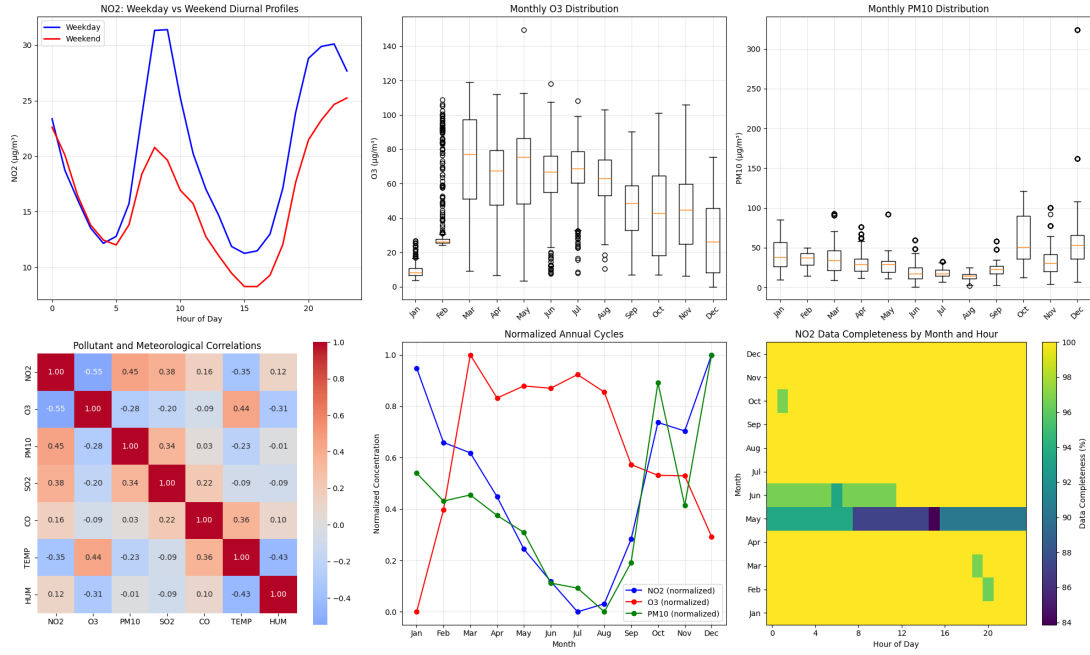


Figure 4: Diurnal NO2 profiles (weekday vs weekend), monthly O3 distribution, monthly PM10 distribution, pollutant-meteorology correlations, normalized annual cycles, NO2 data completeness by month and hour.

4 Meteorological Analysis

4.1 Climatic Parameters

Table 6: Annual meteorological parameters

Parameter	Value
Mean temperature	16.9 °C
Annual thermal range	26.2 °C
Daily thermal amplitude	4.9 °C
Mean relative humidity	77.1 %
Mean wind speed	2.16 m s ⁻¹
Dominant wind direction	WNW (19.7 %)

Temperature ranges from 5.2°C to 31.5°C, with an average diurnal amplitude of 4.9°C. Humidity is high, averaging 77.1%, potentially favoring secondary particle formation.

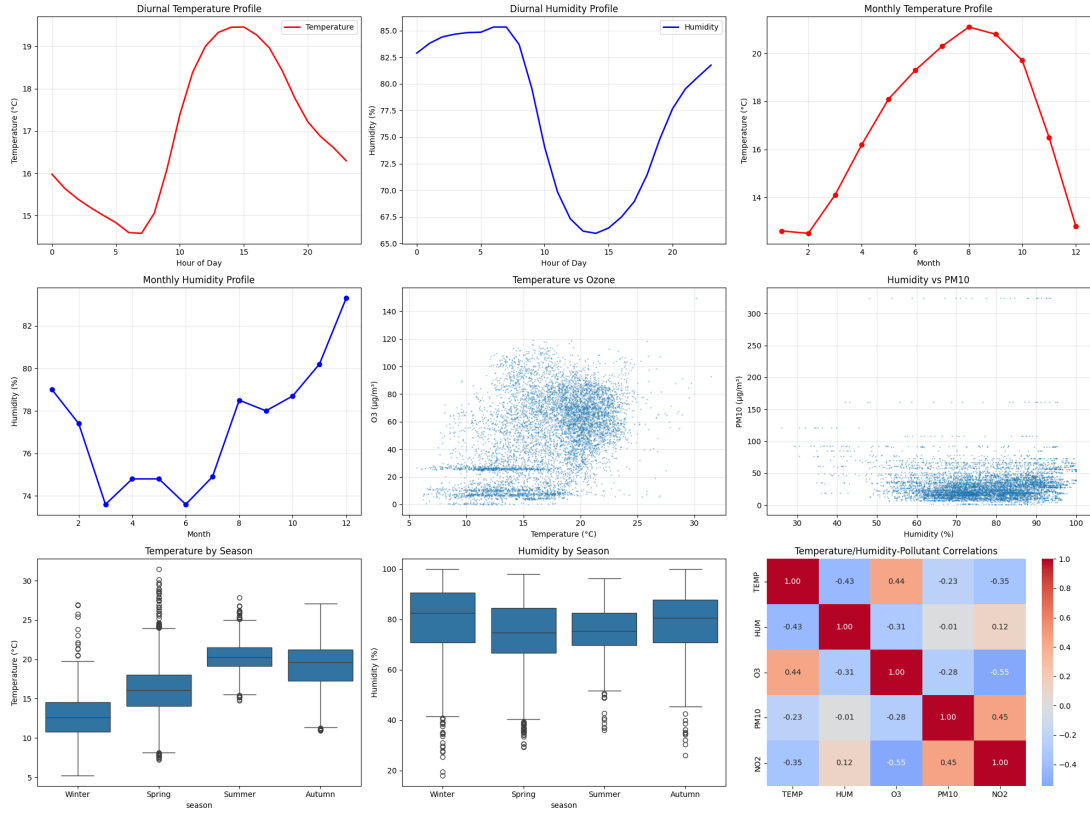


Figure 5: Diurnal temperature and humidity profiles, monthly temperature and humidity profiles, temperature vs ozone, humidity vs PM10, seasonal temperature, seasonal humidity, temperature/humidity/pollutant correlations.

4.2 Pollution-Meteorology Relationships

Table 7: Pollution-temperature-humidity correlations

Pollutant	Temperature Correlation	Humidity Correlation
O ₃	+0.440 (strong)	-0.314 (moderate)
PM ₁₀	-0.232 (weak)	-0.015 (none)
NO ₂	-0.353 (moderate)	+0.123 (weak)
SO ₂	-0.093 (none)	-0.094 (none)
CO	+0.357 (moderate)	+0.100 (weak)

Ozone shows a strong positive correlation with temperature, indicating increased photochemical production in summer. Nocturnal inversions (average strength 1.8°C) increase NO₂ concentrations by 17% at night compared to daytime.

5 Wind Rose Analysis

5.1 Source Identification

Table 8: Wind sectors with high pollution

Pollutant	Dominant Sector	Concentration
NO ₂	North	31.32 $\mu\text{g m}^{-3}$
O ₃	Northwest	62.98 $\mu\text{g m}^{-3}$
PM ₁₀	Northeast	45.48 $\mu\text{g m}^{-3}$
SO ₂	Northeast	5.52 $\mu\text{g m}^{-3}$
CO	Northwest	0.93 m^{-3}

High PM₁₀ and SO₂ concentrations originate primarily from the Northeast, suggesting industrial or dust sources. Calm winds (<1 m/s) increase local concentrations for most pollutants.

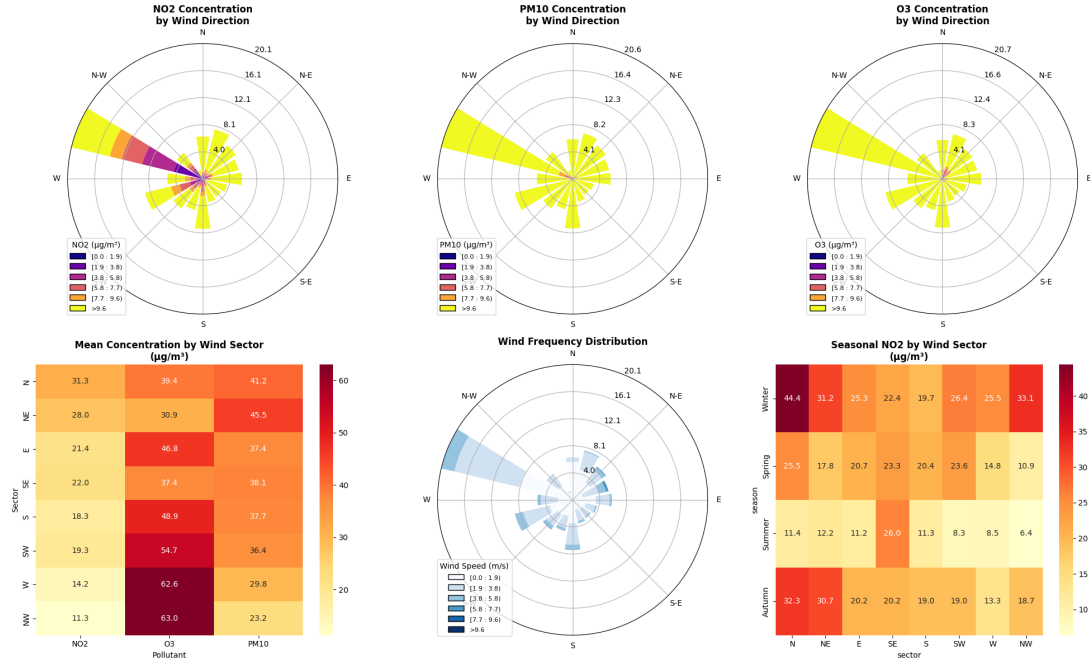


Figure 6: NO₂, PM₁₀, O₃ concentrations by wind direction; average concentrations by wind sector, wind frequency distribution, seasonal NO₂ by sector.

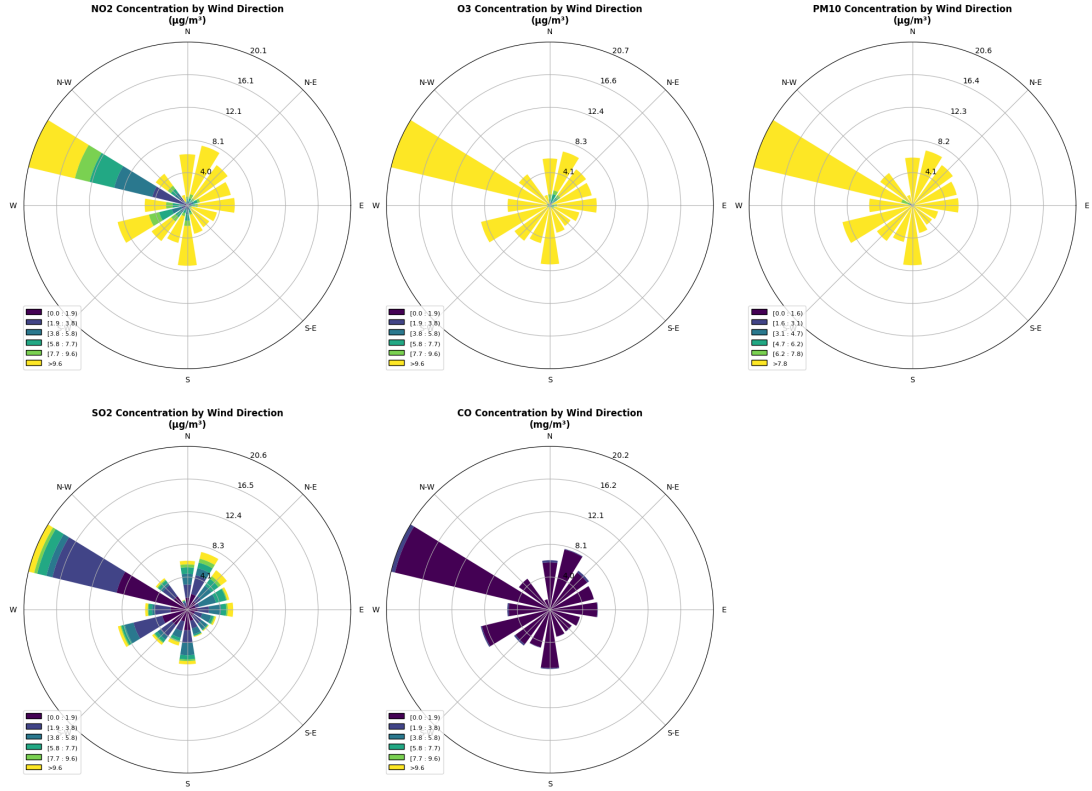


Figure 7: NO₂, O₃, PM₁₀, SO₂, CO concentrations by wind direction.

6 Pollution Episodes

6.1 Major Episodes

Table 9: Significant pollution episodes

Pollutant	Date and Time	Concentration
NO ₂	January 8, 2014, 21:00	157.7 $\mu\text{g m}^{-3}$
PM ₁₀	December 12, 2014, 01:00	324.1 $\mu\text{g m}^{-3}$
SO ₂	May 14, 2014, 19:00	83.1 $\mu\text{g m}^{-3}$

These episodes often coincide with calm winds or thermal inversions, amplifying local emissions.

7 Recommendations

7.1 Corrective Measures

1. **PM₁₀**: Strengthen controls on industrial and construction emissions in the North-east sector, especially during calm wind periods.
2. **O₃**: Implement an alert system for high photochemical production periods, particularly in summer with high temperatures.

3. **Monitoring:** Increase monitoring frequency during critical seasons (winter for primary pollutants, summer for ozone).
4. **Source identification:** Investigate specific sources in the North and Northeast sectors using dispersion modeling.

7.2 Monitoring System Improvements

- Integrate meteorological forecasting models to anticipate pollution episodes.
- Expand the sensor network for better spatial resolution and source identification.
- Implement a real-time alert system based on observed correlations.

Conclusion

The analysis reveals a concerning situation regarding particulate matter (PM_{10}) and ozone (O_3) at the Hay Hassani station, with exceedances linked to meteorological conditions and local sources. While primary pollutants (NO_2 , SO_2 , CO) comply with regulatory standards, persistent exceedances of PM_{10} and O_3 require immediate attention and implementation of targeted corrective measures to improve air quality.